

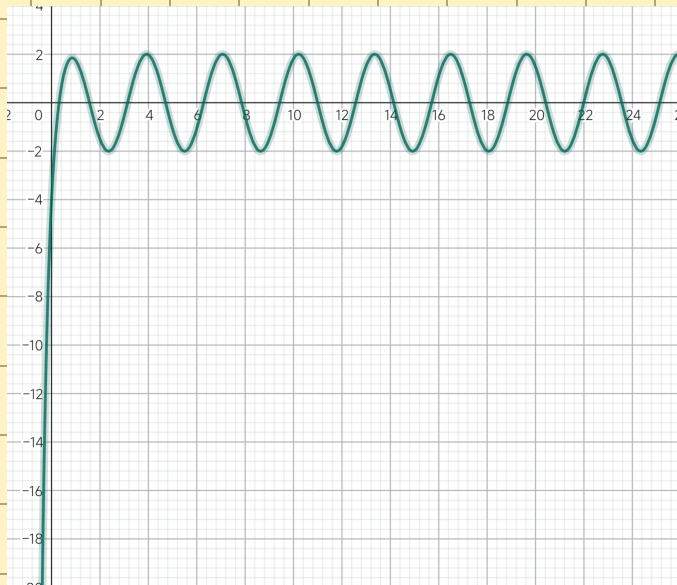
Tarea

$$x(t) = e^{-4t} \quad y(t) = \cos(2t)$$

$$1) z_1(t) = \frac{d}{dt} (x(t) - y(t)) = \frac{d}{dt} (e^{-4t} - \cos(2t))$$

$$\frac{d}{dt} e^{-4t} = -4 e^{-4t} \quad \frac{d}{dt} \cos(2t) = -2 \sin(2t)$$

$$z_1(t) = (-4 e^{-4t}) - (-2 \sin(2t)) = 2(-2 e^{-4t} + \sin(2t))$$



$$2) z_2(t) = \int_0^t (x(\tau)) (y(\tau)) d\tau = \int_0^t (e^{-4\tau}) (\cos(2\tau)) d\tau$$

$$v = \cos(2\tau) \rightarrow dv = -2 \sin(2\tau) d\tau$$

$$dv = e^{-4\tau} d\tau \rightarrow v = -\frac{1}{4} e^{-4\tau}$$

$$I = (\cos(2\tau)) \left(-\frac{1}{4} e^{-4\tau}\right) - \int \left(-\frac{1}{4} e^{-4\tau}\right) (-2 \sin(2\tau)) d\tau$$

$$= -\frac{1}{4} e^{-4\tau} \cos(2\tau) - \frac{1}{2} \int e^{-4\tau} \sin(2\tau) d\tau = I$$

$$\rightarrow v = \sin(2\tau) \rightarrow dv = 2 \cos(2\tau) d\tau$$

$$dv = e^{-4\tau} d\tau \rightarrow v = -\frac{1}{4} e^{-4\tau}$$

$$\int e^{-4\tau} \sin(2\tau) d\tau = -\frac{1}{4} e^{-4\tau} \sin(2\tau) - \int \left(-\frac{1}{4} e^{-4\tau}\right) (2 \cos(2\tau)) d\tau$$

$$= -\frac{1}{4} e^{-4\tau} \sin(2\tau) + \frac{1}{2} \int e^{-4\tau} \cos(2\tau) d\tau \rightarrow I$$

$$\int e^{-4\tau} \sin(2\tau) d\tau = -\frac{1}{4} e^{-4\tau} \sin(2\tau) + \frac{1}{2} I$$

$$I = -\frac{1}{4} e^{-4\tau} \cos(2\tau) - \frac{1}{2} \left[ -\frac{1}{4} e^{-4\tau} \sin(2\tau) + \frac{1}{2} I \right]$$

$$= -\frac{1}{4} e^{-4\tau} \cos(2\tau) + \frac{1}{8} e^{-4\tau} \sin(2\tau) - \frac{1}{4} I$$

$$= \frac{e^{-4\tau}}{40} (-2 \cos(2\tau) + \sin(2\tau))$$

